GraphKAN AMO NQS

Stage 4F-R Final Supervisory Validation Handoff

Frozen 960-run boundary-aware Rydberg NQS campaign

CAMPAIGN + INTEGRITY: PASS | SECOND-HOST REPLAY: OPEN

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Executive statement

All 960 registered work items are COMPLETE: 840 core-lane model runs and 120 RBM control runs. Strict bundle, checkpoint, manifest, registry, and filesystem reconciliation checked all 960 canonical bundles and returned PASS with zero blocking errors, zero filesystem mismatches, and zero duplicate registry keys. The execution source tree still matches all 105 hashes frozen at campaign initialization.

Scientific health is mixed rather than universally green: 831/960 runs pass the exact-health screen and 129 are flagged; 820/960 pass the sampled-health screen and 140 are flagged. These registered failures are retained in every full-grid table and reported as a failure-map result, not removed as invalid files.

The four-phase SIGKILL microcampaign passed and the frozen science roster was not changed after launch. The existing four-run replay also passes technically, but its recorded worker provenance is this same Mac; therefore this package does not claim a genuinely independent second-host replay.

Decision requested

CONDITIONAL GO: accept Stage 4F.1c and Stage 4F-R campaign integrity; hold public release/next-stage escalation until second-host replay is supplied or waived by the supervisor.

1. Acceptance-gate matrix

Gate	Evidence	Status
Typed fail-closed bundles	960 strict bundle validations	PASS
Real checkpoint loading	960 NPZ checkpoints deserialised	PASS
Registry/filesystem reconciliation	0 mismatches	PASS
Protocol/source identity	105/105 frozen hashes match	PASS
Heartbeat + attempt fencing	Kill/retry evidence and terminal registry	PASS
Four planned process kills	training/checkpoint/promotion/post-promotion	PASS
16-run real microcampaign	16/16 COMPLETE	PASS
Frozen Stage 4F-R campaign	960/960 COMPLETE	PASS
Exact scientific health	831 VALID; 129 FLAGGED	MIXED
Sampled scientific health	820 VALID; 140 FLAGGED	MIXED
Independent second-host replay	4/4 PASS, but same-host provenance	OPEN
Public artifact release	Await supervisor decision on replay	HOLD

Interpretation

The engineering closure and full campaign integrity gates pass. Final external reproducibility remains a distinct condition and is not inferred from same-host execution.

2. Frozen protocol and dataset integrity

Item	Frozen value
Campaign ID	stage4f_r_frozen__stage4f-native-v1c__P327f66318ee93c33
Protocol SHA-256	327f66318ee93c3365f5a12ba4d64fddd20a29ee288cc30330d437f514328ff9
System	N=10 Rydberg chain; omega=1.0
Registered grid	2 boundaries x 4 delta x 3 C6 x 8 models x 5 seeds = 960
Primary outcomes	Exact energy, density, connected correlation
Sampling	4096 samples; SR shift 0.1
Canonical bundles	960
Quarantined late attempts	35 retry events retained as noncanonical evidence
Source ledger	105 entries; 0 mismatches

Validation boundary

The canonical campaign directory is treated as immutable. All tables and figures in this handoff were written outside the frozen directory. Analysis operates only on canonical COMPLETE bundles; quarantined late attempts are retained for engineering audit but excluded from science tables.

3. Preregistered exact outcomes

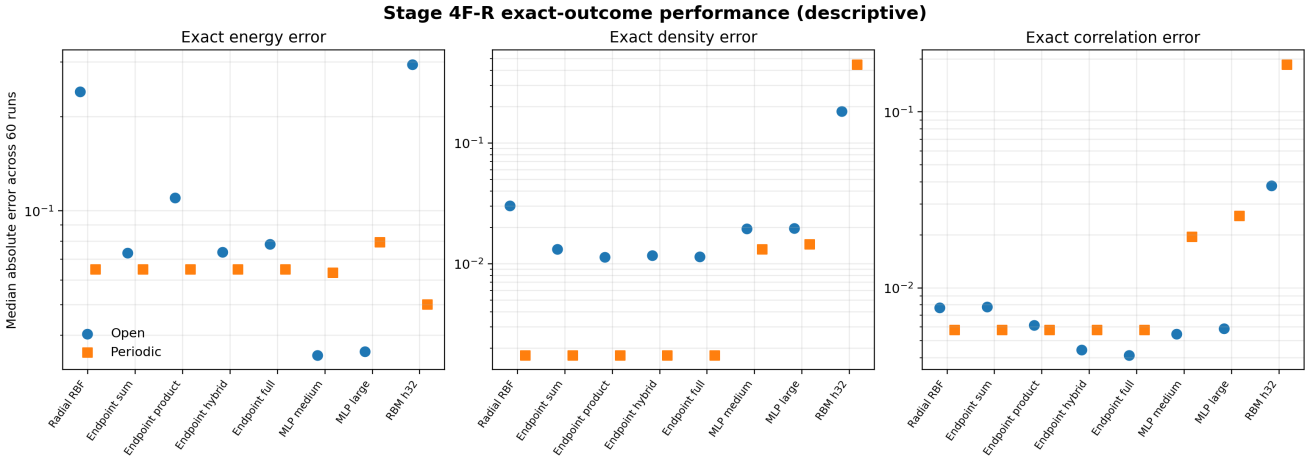


Figure 1. Median absolute error across registered runs, separated by boundary. Log scale; descriptive only.

Boundary	Outcome	Lowest descriptive median	Median error
Open	Energy	MLP medium	0.0345
Open	Density	Endpoint product	0.01129
Open	Correlation	Endpoint full	0.004145
Periodic	Energy	RBM h32	0.05023
Periodic	Density	Endpoint full	0.001746
Periodic	Correlation	Endpoint full	0.005759

Health screen: 129 exact-health and 140 sampled-health rows are flagged. The complete registered grid, including those failures, is used in the displayed medians. These rankings do not establish generic model superiority and must be interpreted jointly across all three outcomes, parameter count, runtime, and boundary.

4. Boundary-conditioned endpoint mechanism

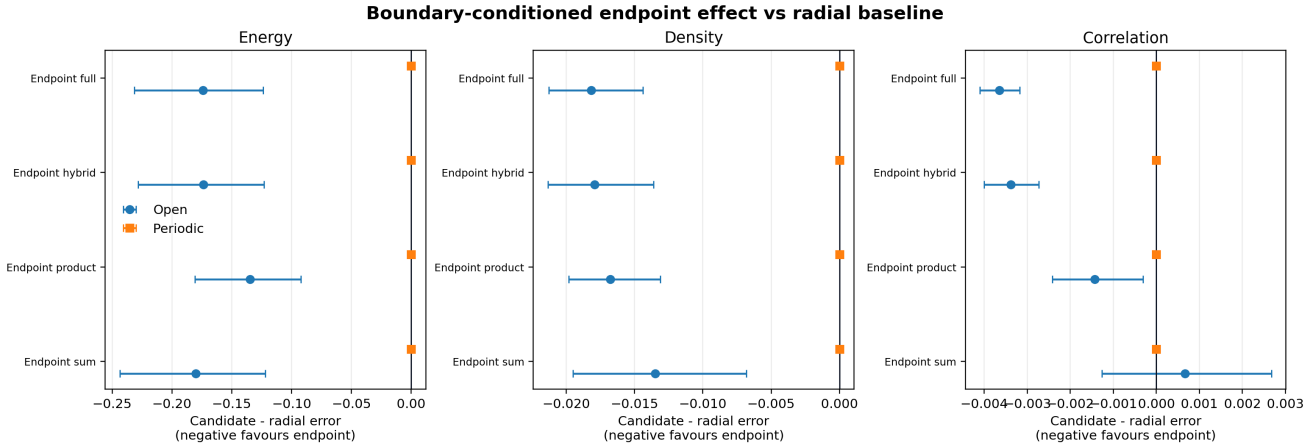


Figure 2. Cell-level paired endpoint-minus-radial error. Negative favours the endpoint variant. Bars are deterministic 95% bootstrap intervals over registered physics cells after median reduction across five seeds.

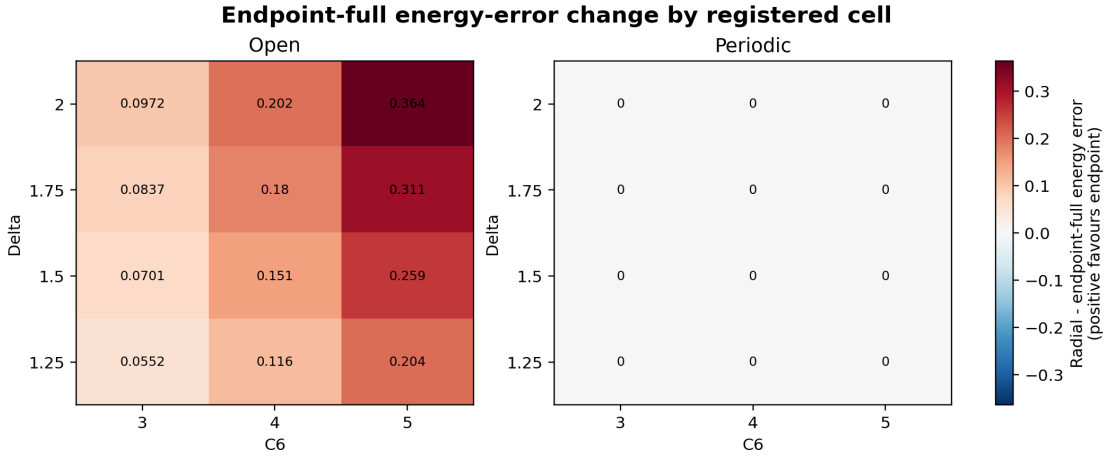


Figure 3. Registered-cell map for endpoint-full versus radial exact-energy error. Positive values favour endpoint-full.

5. Accuracy-runtime trade-off

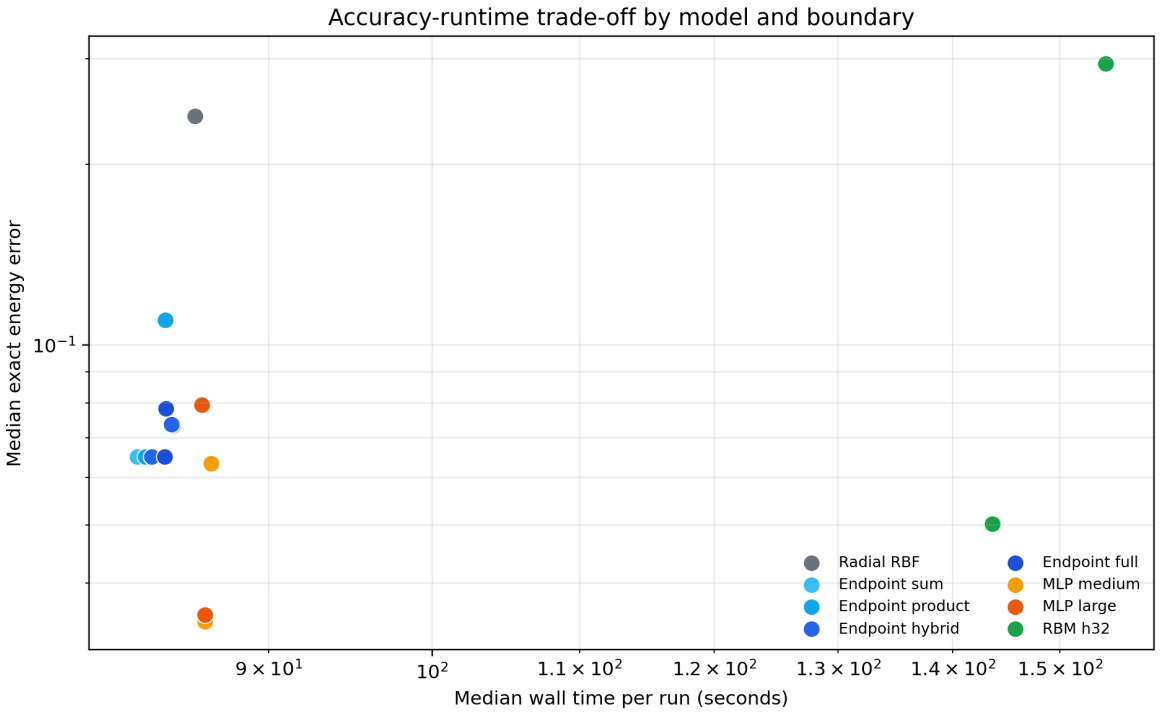


Figure 4. Each model contributes one open and one periodic aggregate point. Both axes are logarithmic.

Statistical handling

The registered physics cell is the comparison unit. Five seeds are first reduced by their cell median. Paired candidate-minus-reference comparisons report mean and median differences, candidate win fraction, deterministic bootstrap intervals, Wilcoxon signed-rank p-values, and Holm adjustment within each scope/outcome family. These outputs are supplied for supervisory analysis and are not converted into a universal-superiority claim.

6. Crash recovery and retry evidence

Fault phase	Lane	SIGKILL observed	Recovery exit	Gate
during-training	core	True	0	PASS
during-checkpoint	core	True	0	PASS
during-promotion	control	True	0	PASS
after-promotion	core	True	0	PASS

The 16-run microcampaign terminated 16/16 COMPLETE after four deliberate process kills. The full Stage 4F-R registry records all 960 rows COMPLETE. Late or lease-lost attempts were quarantined and did not become canonical.

Replay disclosure

The four-run replay validates 4/4 canonical bundles with zero reconciliation errors. Its worker IDs contain 'Subhojits-MacBook-Air', so it is a clean same-host replay, not independent second-host evidence. The package therefore requests either a genuine separate-host replay or an explicit supervisory waiver before public release.

7. Reproducibility and review commands

```
cd reports/stage4fr_final_20260721 && shasum -a 256 -c SHA256SUMS.txt && cd ../..

PYTHONPATH="$PWD/src" python scripts/stage4f1c_campaign.py validate --campaign results_stage4f_r/frozen --strict-integrity
--reconcile-registry --require-terminal --require-complete

python -m pytest -q

python scripts/finalize_stage4fr_publication.py --campaign results_stage4f_r/frozen --analysis-output reports/stage4fr_final_20260721/analysis
--report output/pdf/GraphKAN_Stage4FR_Final_Validation_Report_20260721.pdf --skip-package
```

Reviewer checklist

- Confirm the regenerated report checksum ledger.
- Re-run strict campaign validation; require 960/960 COMPLETE and PASS.
- Confirm campaign protocol hash 327f66318ee93c3365f5a12ba4d64fddd20a29ee288cc30330d437f514328ff9.
- Inspect kill_microcampaign_summary.json for four SIGKILL/recovery passes.
- Review analysis tables and figures without promoting descriptive rankings to universal claims.
- Decide whether a genuine second-host replay is required before next-stage/public release.

Requested decision

CONDITIONAL GO: accept Stage 4F.1c and Stage 4F-R campaign integrity; hold public release/next-stage escalation until second-host replay is supplied or waived by the supervisor.